

STORAGE – SSDs

+1 Year, +1000 MBps. SSDs are really pushing the envelope

As we were pondering over last year's Zero1 Awards, we'd expected NVMe to achieve mainstream popularity with cheaper SKUs compared to last year. While things have gotten a lot cheaper than last year, it still isn't within the affordable price segment for the average customer. Even now, most of the SSD sales in the market are dominated by TLC NAND which aren't the cream of the crop. You can get individual SKUs for about ₹3,000 for a 120 GB drive and it's a quick and easy way to join the SSD revolution. However, one of the significant changes this year was the advent of hybrid SSDs. Let us clarify, these are not a combination of traditional hard drives with a small capacity SSD NAND for caching. That's what hybrids mean when you're talking storage these days. The new trend that we are speaking of is when manufacturers combine SLC+MLC or MLC+TLC NAND in one drive. While this does present certain wonderful possibilities for managing wear on the data cells, it also throws a spanner into a little thing called benchmarking.

So what's new? There's 3D XPoint which is the next big thing for memory standards. Intel and Micron have already demonstrated prototypes of the new technology. So while next year might be a bit uneventful till 3D XPoint drives hit the consumer market, things have definitely improved this year.

ZERO1 WINNER

Samsung 960 Pro 1 TB NVMe SSD

Everything inside the Samsung 960 Pro is an improved variant of what the 850 Pro had. The UBX controller has been replaced by the new

Polaris lineup and the NAND while being the same 3D V-NAND technology, now packs in a lot more transistor layers. There's a 100 per cent gain in density which has resulted in an equally tremendous gain in IOPS. 4K Random Read at a Queue Depth of 32 has improved by approximately 50 percent over the 950 Pro.

All of these changes make the 960 Pro a worthy upgrade for enthusiasts as you can now get up to 2 TB of storage space while enjoying read speeds of 3,500 MB/s and write speeds of 1,200 MB/s. That's roughly 1,000 MB/s greater than last year's winner of the Zero1 Awards. Oh! And this flagship does sport a price tag to match.



**ZOTAC SONIX
GAMING SSD 480 GB**
Price: ₹33,799



**Samsung SM951
512 GB**
Price: ₹23,199



**Samsung 850 EVO
250 GB**
Price: ₹7,499

RUNNER-UPS

ZOTAC SONIX GAMING SSD 480 GB, Samsung SM951 512 GB

The SONIX SSD sports read speeds of 2,600 MBps while the 10th anniversary refresh offers about 2,800 MBps read speeds. Since we haven't received the units we don't know if the internals do indeed differ or if the improvement is purely firmware based. Regardless, it was in the running for the Zero1 Award until the 960 Pro showed up on our doorstep.

Then there's the Samsung SM951 which did release last year globally but wasn't available in India till much later. In fact, we sourced our unit from one of the numerous other products that came in for the Zero1 awards. Even the SM961 which released around June this year is yet to arrive in India. The SM951 trails behind the SONIX with sequential read speeds of 2,150 MB/s, thus, bagging the third place.

BEST BUY

Samsung 850 EVO 250 GB

It's gotta get a little tiring to see another Samsung SSD in this list but that's purely because the 850 EVO really fits the bill. While the 850 EVO does have the same controller as the 850 Pro, which happens to be Samsung's own MHX controller. The key difference is the NAND, the 850 EVO uses TLC NAND and the 850 Pro opts for the more durable MLC NAND. This is why we see the 850 Pro coming with a warranty period of 10 years while the 850 EVO comes with exactly half of that. A combination of great durability, excellent IOPS performance nearly equaling the 850 Pro and good pricing have led to the 850 EVO bagging the Best Buy. And it's not just us, the 850 EVO pretty much tops the list of most popular SSDs across the globe for these very reasons. 

Samsung 960 Pro 1 TB NVMe SSD

Price: \$629 (₹43,000+Taxes)

